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A ligand-centered framework for $\gamma\delta$ T cell activation in colorectal cancer revealed by single-cell and transformer-based perturbation

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Understanding the activation mechanisms of $\gamma\delta$ T cells in colorectal cancer (CRC) is critical for harnessing their therapeutic potential. Here, using an atlas of human CRC-infiltrating $\gamma\delta$ T cells that we built by integrating multiple single-cell RNA-seq datasets, we developed a $\gamma\delta$ T cell-refined ligand inference pipeline by combining differential gene expression, gene regulatory network prediction, ligand inference, and in silico perturbation analysis. This approach identified ligands, including IL-15 and TNFSF9 (4-1BBL), as candidates promoting $\gamma\delta$ T cell effector function and highlighted NCR2 and KLRC3 (NKG2E), whose in silico overexpression was associated with $\gamma\delta$ T cell activation. Ligand enrichment analyses further indicated that monocytes and dendritic cells are key contributors to $\gamma\delta$ T cell activation in the tumor microenvironment. Our results also highlighted transcription factors IKZF1, FOSL2, and FOXO1 in the less activated $\gamma\delta$ T cells and IRF1, KLF2, and BHLHE40 in the effector $\gamma\delta$ T cells that plausibly regulated the differential activation state. Together, our results offer a systems-level view of the signaling and transcriptional programs governing $\gamma\delta$ T cell phenotypes in CRC and provide a foundation for $\gamma\delta$ T cell-based immunotherapies with enhanced antitumor functions.

KEYWORDS

cancer colorectal, deep learn ing, gamma delta ($\gamma\delta$) T cells, ligand, perturbation prediction, transformer

Introduction

Colorectal cancer (CRC) is one of the most frequently diagnosed cancers and the leading cause of cancer-related deaths (1–3). It originates in the colon or rectum and is characterized by the abnormal proliferation of epithelial cells (4). Surgery is the primary treatment for advanced resectable CRC, yet recurrence occurs in a substantial proportion of patients with stage II or III disease (5, 6). Most recurrences occur within 5 years after

surgery, and the relapse significantly increases the risk of metastasis and decreases the survival rate (5, 7, 8). For non-resectable CRC, radiotherapy and chemotherapy are standard-of-care therapies that can reduce tumor burden (9, 10) but are limited by systemic toxicity, adverse effects, and a lack of tumor specificity (11). A substantial proportion of patients eventually develop incurable recurrent disease, even after standard treatments (11). Therefore, there is a need for novel treatments that can be used in addition to conventional treatments to prevent recurrence.

$\gamma\delta$ T cells are a unique subset of immune cells with potent anti-tumoral ability (12, 13). They can be activated independently of classical MHC-mediated antigen presentation and, in some cases, through TCR-independent mechanisms involving the engagement of stress-induced ligands on target cells (14). Although $\gamma\delta$ T cells account for less than 5% of human blood T lymphocytes, their proportion is significantly higher in the colon mucosa (15). They account for 10–40% of T cells in the epithelium monolayer (16–18), where CRC oncogenesis usually begins (19). These cells exhibit dynamic surveillance behavior, migrating through the intestinal epithelium in a flossing-like manner and traversing the basement membrane to access the lamina propria (15, 20). $\gamma\delta$ T cells have been shown to inhibit intestinal tumor growth via CD103-E-cadherin interactions, highlighting their role in maintaining epithelial integrity and tumor surveillance under homeostatic conditions (21). Therefore, they are promising candidates for immunotherapeutic strategies because they are naturally present in the tumor microenvironment, exhibit strong tissue infiltration, and recognize targets in an MHC-independent manner, enabling allogeneic transfer.

However, the mechanism underlying the activation of $\gamma\delta$ T cells in CRC remains unclear. In our previous study, we integrated single-cell RNA sequencing (scRNA-seq) data from multiple CRC studies and revealed diverse states among CRC-infiltrating $\gamma\delta$ T cells. These include a substantial population of unexhausted, tissue-resident bystanders expressing *ITGAE* and *ITGA1*, as well as highly cytotoxic effector cells expressing *IFNG* (22). We therefore hypothesize that identifying the molecular cues driving the activation of these bystanders could support tumor cell elimination and reduce CRC recurrence, given that recurrence often arises from residual tumor cells (7).

It is well established that intraepithelial $\gamma\delta$ T cells interact closely with colon epithelial cells under homeostatic conditions (21). Through both T cell receptor (TCR)-dependent and TCR-independent pathways, they detect ligands expressed on epithelial cells or present in the local microenvironment. T cell activation also depends on the co-stimulatory ligands they engage and the cytokines they receive. Taken together, these insights support the hypothesis that $\gamma\delta$ T cells may undergo a shift from a less activated to a more activated state through TCR-independent ligand-receptor interactions.

Single-cell RNA sequencing is transforming CRC research by enabling high-resolution profiling of gene expression in heterogeneous tumor tissues, capturing the activity of individual cell types, including tumor-infiltrating immune cells. Here, we combine our previous multi-study integration of $\gamma\delta$ T cell scRNA-seq data with gene regulatory network (GRN) inference and large language model (LLM)-

based in silico gene perturbation to uncover the molecular drivers of $\gamma\delta$ T cell activation in CRC. Through this integrative approach, we identified genes that correlate with, and may potentially drive, the activation transition, and made backward inferences about upstream ligands regulating them via signaling pathways.

Result

CRC-infiltrating $\gamma\delta$ T subsets have varied activation levels

In our previous study, we integrated $\gamma\delta$ T cells from multiple human CRC datasets to construct a comprehensive cell atlas with over 18,000 $\gamma\delta$ T cells, of which 9,201 were identified as isolated from tumors by the data providers, across diverse anatomical sites, patient demographics, and disease stages, providing a broad view of $\gamma\delta$ T cell transcriptional states in CRC (22) (Figure 1a). We performed clustering on this integrated dataset and identified $\gamma\delta$ T cell subtypes based on canonical $\alpha\beta$ T cell functional subset marker genes curated from the literature, including effector T cells (Teff), tissue-resident memory T cells (TRM), progenitor exhausted T cells (Tpex), and exhausted T cells (Tex), all present in both tumor and adjacent normal tissues, with Teff and Tex more enriched in tumors. Notably, although we use the term “exhausted” to describe $\gamma\delta$ T cells that express canonical exhaustion markers, emerging evidence indicates these cells may not be functionally impaired (23, 24). A more accurate descriptor is therefore “exhaustion-marker-expressing” $\gamma\delta$ T cells. For brevity, however, we will continue to refer to this population as “Tex” throughout the manuscript. We observed that the TRM-like population, the major subtype among tumor-infiltrating $\gamma\delta$ T cells in our integrated dataset (Figure 1b), did not highly express conventional exhaustion markers such as *PDCD1* and *TIGIT* or effector-related genes including *GZMB*, *NKG7*, *PRF1*, or *IFNG* (Figure 1c). Gene set scoring based on effector genes (*IL2RA*, *CD38*, *NKG7*, *GZMB*, *PRF1*, *IFNG*) and exhaustion genes (*PDCD1*, *CTLA4*, *LAG3*, *HAVCR2*, *TIGIT*) further confirmed that the TRM-like population exhibited the lowest scores for both effector and exhaustion signatures (Figure 1d).

Given the plasticity of T cells, we aimed to identify candidate ligands that could drive the potential transition from TRM-like bystanders to Teff cells in CRC. Using scRNA-seq data, we developed a pipeline that integrates multiple computational tools to perform ligand inference (Figure 2). At the core of this pipeline is NicheNet (25), a computational framework that performs a random walk on a ligand to target gene transcription network that is constructed from known ligand-receptor interactions, intracellular signaling, and gene regulatory networks to infer the ligands that are most influential in regulating a given set of genes. For the standard processing, we input differentially upregulated genes in the Teff compared to TRM as the signature associated with the state transition into NicheNet, which will output the ligands that most influence these differential expression events. We chose NicheNet primarily because it directly links ligands to downstream target gene

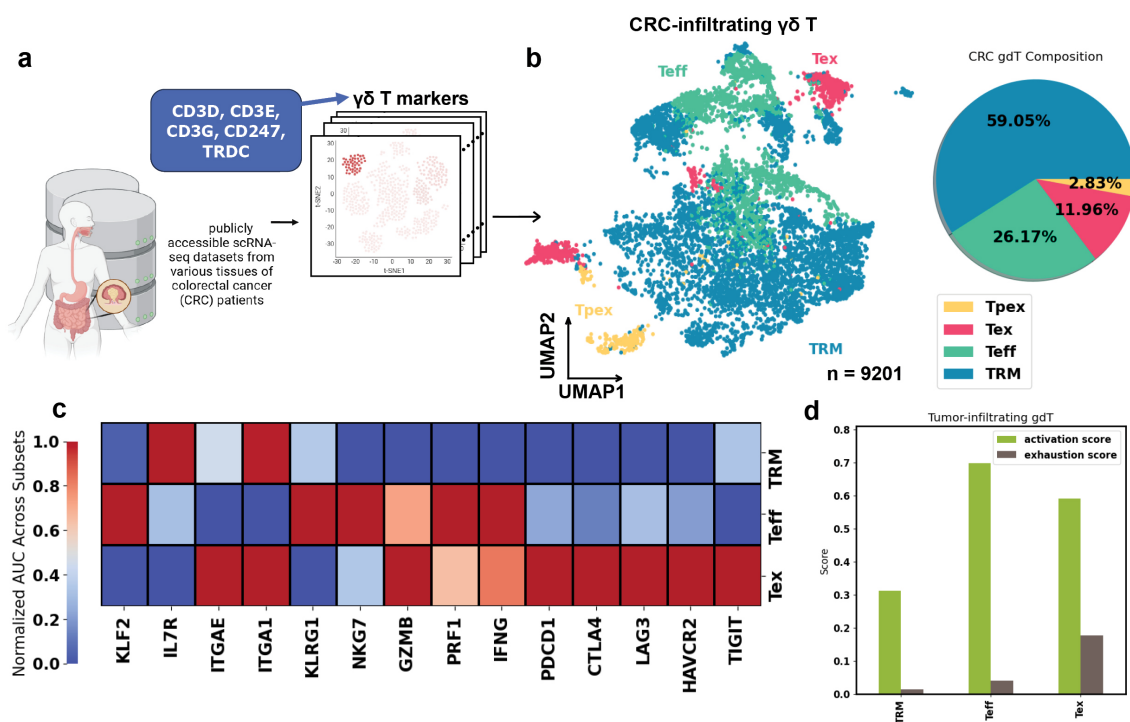


FIGURE 1

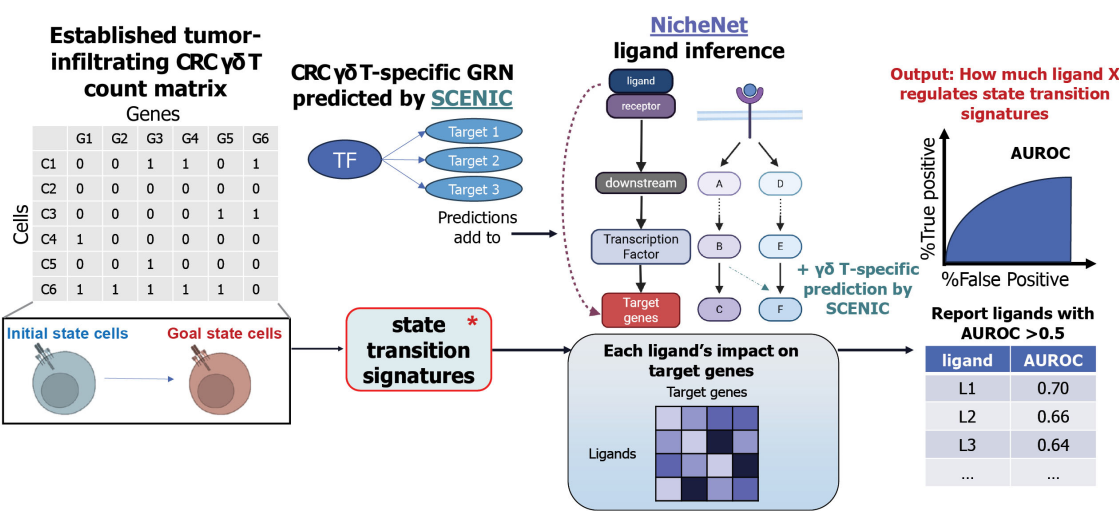
Overview of the heterogeneous human CRC $\gamma\delta$ T cells. **(a)** Using the minimum set of $\gamma\delta$ T cell transcriptomics markers we developed, we examined publicly available CRC scRNA-seq datasets and identified $\gamma\delta$ T clusters. They are integrated to build a human CRC $\gamma\delta$ T cells dataset. **(b)** (Left) Uniform manifold approximation and projection (UMAP) of $\gamma\delta$ T cells showing subsets: Teff (effector-like T cells), TRM (tissue-resident memory T cells), Tpex (progenitor exhausted T cells), and Tex (exhausted T cells). (Right) Composition of $\gamma\delta$ T cell subtypes in the tumor. **(c)** Heatmap showing the T cell function-related marker RNA expression in TRM, Teff, and Tex. Expression has been normalized across cell types. **(d)** Bar plot showing the effector activation score and exhaustion score in $\gamma\delta$ T cell subsets.

transcription, rather than stopping at ligand-receptor co-expression. In addition, compared with other network-based approaches, NicheNet is explicitly receiver-centric, modeling how putative ligands affect the transcriptional programs of a specified receiver population. This receiver-focused design is particularly advantageous for rare cell types like CRC-infiltrating $\gamma\delta$ T cells, where the identity of the dominant sender populations is not always obvious *a priori* in the complex TME. It also provides flexibility to interpret ligand sources and to prioritize potential experimental interventions on either ligands or receptors in follow-up work.

One challenge in this style of ligand inference is that the differentially expressed genes (DEGs) between TRM and Teff $\gamma\delta$ T cells identify genes associated with their transcriptional differences, but such differences may reflect a mixture of upstream, downstream, and unrelated processes. To prioritize gene programs more likely to contribute to the transition between these states, we incorporated Geneformer (26), a large language model pre-trained on human scRNA-seq data for *in silico* perturbation analysis, into our pipeline to identify genes whose *in silico* overexpression is associated with a shift from a TRM-like to a Teff-like transcriptional profile. In our modified workflow, NicheNet was used to calculate another set of ligands that can regulate Geneformer-predicted gene expression events, leading to cell state transition. We chose Geneformer because it is pretrained exclusively on a large corpus of human single-cell data and provides a built-in perturbation function that directly links simulated changes in

gene expression to shifts in the cell-state embedding, quantified using cosine similarity. Other large pretrained models such as scGPT (17) or scFoundation (18) can generate binned or continuous perturbed gene expression profiles, and in principle one could derive state embeddings and perform Geneformer-like quantification from those outputs. However, doing so would require customization of their original pipelines. In contrast, Geneformer offered a robust and immediately applicable framework for ranking 'state-shifting' genes in our TRM-to-Teff transition analysis, while we view alternative models as promising avenues for future extensions.

A further challenge in ligand signaling inference is that gene regulatory networks (GRN) are cell type-specific (27) and $\gamma\delta$ T cells are a less well-studied population whose cellular programs may not be well-represented in prior knowledge networks. To address this limitation of existing workflows, we integrated ligand-receptor interactions relevant to $\gamma\delta$ T cells, curated from the literature (14, 28–33), with SCENIC (34)-inferred transcription factor (TF)-target gene relationships into the NicheNet current default network. Using this $\gamma\delta$ T cell-tailored network, we performed finely tuned $\gamma\delta$ T cell-specific ligand inference to prioritize candidate ligands and gene expression programs associated with $\gamma\delta$ T cell activation in CRC. We chose SCENIC for GRN inference primarily because it balances co-expression analysis with transcription factor motif enrichment, which is particularly advantageous when prior knowledge about a rare cell type like $\gamma\delta$ T cells is limited. Unlike approaches that rely solely on



***To define the “state transition signatures”:**

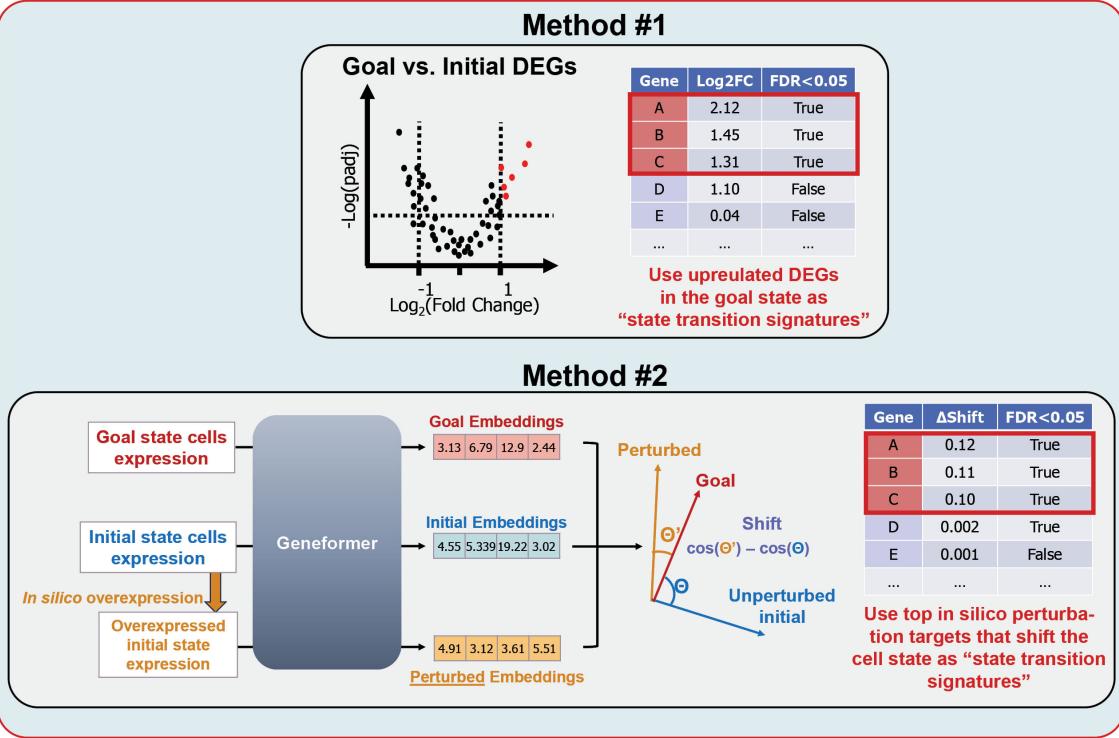


FIGURE 2
 $\gamma\delta$ T cell-refined ligand inference pipeline. Overview of the computational pipeline. We first define a potential cell state transition from an initial state (i.e., Teff) to a goal state (i.e., TRM) in the scRNA-seq count matrix. Then, we use NicheNet, a ligand inference tool that uses Personalized Page Rank on the constructed ligand to target gene network, known networks summarized by NicheNet authors from databases, and $\gamma\delta$ T-specific networks predicted by SCENIC based on the co-expression of known TF with potential target genes in our data, to approximate the propagation of the signal in order to compute each ligand’s regulatory power on its target gene transcription. By computing the AUROC between the transition-associated signatures and each ligand to target genes’ regulatory potential vector, we can infer which ligand has the highest potential of regulating the given signatures. The transition signatures are calculated in two ways: differential gene expression analysis and in silico perturbation analysis using Geneformer, a language model. For visualization, the embedding is drawn as a 4-element array here as a schematic representation of a 512-dimensional vector.

enhancer-gene correlations [for example GRaIE (12) or ANANSE (13)] or on sparse regression [for example Inferelator (14)], SCENIC combines intuitive tree-based gene-expression modeling with motif enrichment analysis around target gene promoters. This allows relatively robust identification of high-confidence TF-target interactions even in the absence of clear condition contrasts or additional omics layers such as ATAC-seq [required by methods like sc-compReg (15)] or lineage information [used by approaches such as scMTNI (16)]. This biologically interpretable framework, together with SCENIC’s broad adoption and mature software

ecosystem, made it a practical choice for $\gamma\delta$ T cell-specific GRN inference in our setting.

Integration of SCENIC-predicted novel gene regulatory relationships maintains general predictive performance

The first step was to infer CRC-infiltrating $\gamma\delta$ T cell-specific TF-target gene relationships from the integrated single-cell atlas. Using SCENIC, we identified 30,025 significant and distinct TF-target gene pairs in our $\gamma\delta$ T cell atlas, of which 56% were already documented in the NicheNet default GRN (Figure 3a; full list of regulons in Supplementary Table S1). We used publicly available human T and NK cell ChIP-seq data to assess whether the remaining 44%, approximately 13,200 novel TF-target gene pairs specific to CRC-infiltrating $\gamma\delta$ T cells, could be validated using an orthogonal experimental setup. From ChIP-Atlas (35–37), 19 of the predicted TFs had available peak information, and for 15 of them, ChIP-seq peaks were found within ± 5 kb of the transcription start site (TSS) of at least a subset of their SCENIC-predicted target genes (Figure 3b). Regulon specificity scores across the four $\gamma\delta$ T cell subsets indicated distinct TF regulatory programs, with TRM cells primarily regulated by TFs such as IKZF1 and FOXO1, Teff cells by KLF2 and interferon regulatory factor 1 (IRF1), Tex cells by MYBL2 and SPI1, and Tpex cells by RUNX1, EGR3, and NFKB1 (Figure 3c).

To test whether the SCENIC-inferred cell subtype-specific TFs are functionally linked to cell state, we used CellOracle (38) to simulate *in silico* knockouts (KOs) of the top 5 TFs identified in TRM and Teff cells, since we are mainly interested in promoting the $\gamma\delta$ T cells' effector function. Briefly, CellOracle utilizes reference GRN and then fits regularized linear models to learn context-specific edge weights that predict target-gene expression from candidate TFs. KO effects are simulated by setting the TF of interest to 0 expression, propagating signals through this GRN, and estimating cell-state transition probabilities. Indeed, although with varied effects, it is predicted that by knocking out the top TRM-specific TFs FOXO1, IKZF1, ELF1, and NR3C1, the CRC $\gamma\delta$ T cells shift towards the effector state, which is visualized by the general trend of field vectors flow from the cluster of TRM to the cluster of effectors (Figure 4). Similarly, by knocking out the top Teff-specific TFs IRF1, BHLHE40, KLF2, and ATF3, the CRC $\gamma\delta$ T cells shift towards the TRM state.

All novel TF-target gene pairs were incorporated into the NicheNet gene regulatory network. In the original NicheNet study, model performance was evaluated by predicting ligands from RNA-seq data derived from experimentally perturbed conditions using known ligands, referred to as the “gold standard.” Because $\gamma\delta$ T cell-specific additions primarily expand underrepresented signaling and regulatory interactions rather than altering shared core pathways, we expected the model performance on unrelated test data to remain stable. Indeed, benchmarking confirmed that integrating both the SCENIC-predicted GRN and literature-curated $\gamma\delta$ T cell ligand-receptor interactions did not compromise the model's ability to correctly identify perturbed ligands in the reference dataset (Supplementary Figure S1).

NicheNet combined with SCENIC infers ligands including 4-1BBL and IL-15 that shape $\gamma\delta$ T cell activation in CRC

After customizing the NicheNet model, we computed DEGs between Teff and TRM $\gamma\delta$ T cells in our atlas. In addition to classical cytotoxic molecule-encoding genes such as *GZMB*, *NKG7*, and *GNLY*, and effector function-related genes like *IFNG*, Teff cells also upregulated *KLRG1*, *CX3CR1*, and *ZNF683* (encoding Hobit) (Figure 5a; full list in Supplementary Table S2). Collectively, these DEGs reflected the effector-associated transcriptional profile of Teff cells.

To identify potential upstream regulators of this transcriptional program, we input the Teff-upregulated genes into NicheNet to infer ligands most likely to drive the TRM-to-Teff transition. Among the top 15 predicted ligands were IL-4, IL-15, IL-21, and TNFSF9 (encoding 4-1BBL), suggesting they may contribute to promoting $\gamma\delta$ T cell activation. However, the activation landscape also included immunosuppressive signals, with TGFBI, IL10, and PD-L1 (CD274) predicted as counter-regulatory ligands (Figure 5b; list in Supplementary Table S3). For comparison, we also used the DEG-based NicheNet approach to predict ligands that potentially drive the transition from Teff to Tex. This analysis identified ligands corresponding to several inhibitory receptors, including CD274 (PD-L1, binding PD-1), LGALS9 (binding TIM-3), NECTIN2 (binding TIGIT), and CDH1 (E-cadherin, binding KLRG1), highlighting their potential roles in promoting effector exhaustion (Figures 6a, b; list of DEGs in Supplementary Table S4; list of ligands in Supplementary Table S5).

Although NicheNet evaluates the full list of differentially expressed genes (DEGs) to generate ligand predictions, we asked which target genes were most strongly associated with the top-ranked ligands. Specifically, we examined the top five target genes for each of the top predicted ligands and assessed their expression in the goal cell states, Teff and Tex, to gain insight into the basis of the NicheNet predictions (ligand-target gene matrices in Supplementary Figures S2a, b). Several of the top target genes were indeed highly expressed in goal state cells (Figures 5c, 6c). Among them, *IFNG*, *CCLAL2*, and *IER3* stood out as some of the most upregulated genes in the Teff population (Figure 5a).

We also compared the ligand prediction results of the unmodified NicheNet model with those of the $\gamma\delta$ T cell-tailored NicheNet. The inclusion of $\gamma\delta$ T cell-specific edges did not significantly increase the number of novel ligands predicted to perform better than random (defined as AUROC > 0.5), but it did adjust the relative ranking and confidence of ligand predictions within the $\gamma\delta$ T cell context (Supplementary Figures S3a–c; full list of ligands predicted by unmodified NicheNet in Supplementary Tables S6, S7). This refinement can support future experimental designs by prioritizing ligands that are more likely to have a functional impact on $\gamma\delta$ T cells. Using the annotated CRC whole-tissue scRNA-seq reference dataset from Joanito et al. (39), we further assessed the enrichment of the top 30 $\gamma\delta$ T cell phenotype-shifting ligands across various cell types in the CRC tumor microenvironment. iCMS2-type CRC cells, found in microsatellite-stable (MSS) patients, were enriched for ligands predicted to regulate the TRM-to-Teff transition, whereas iCMS3 cells,

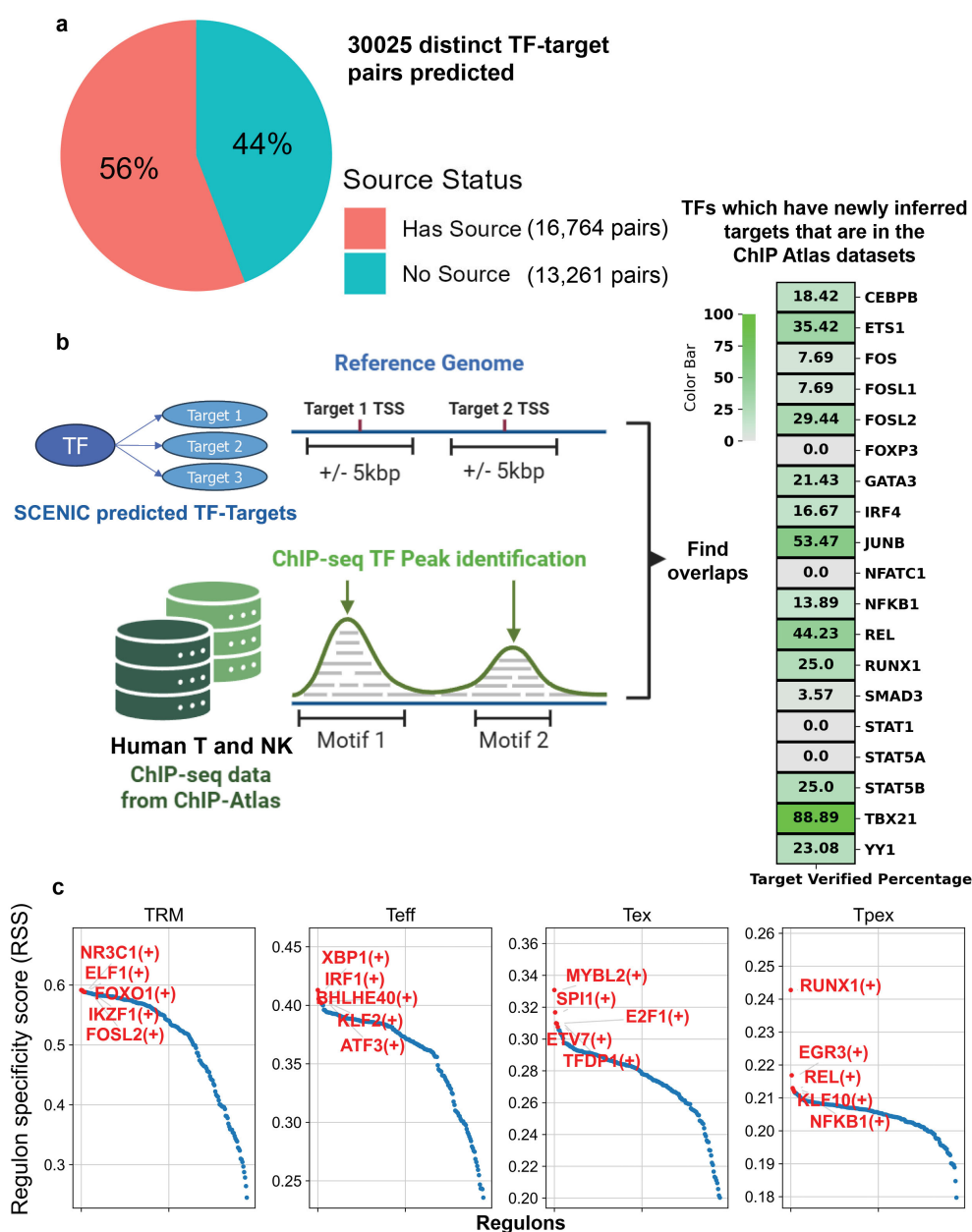
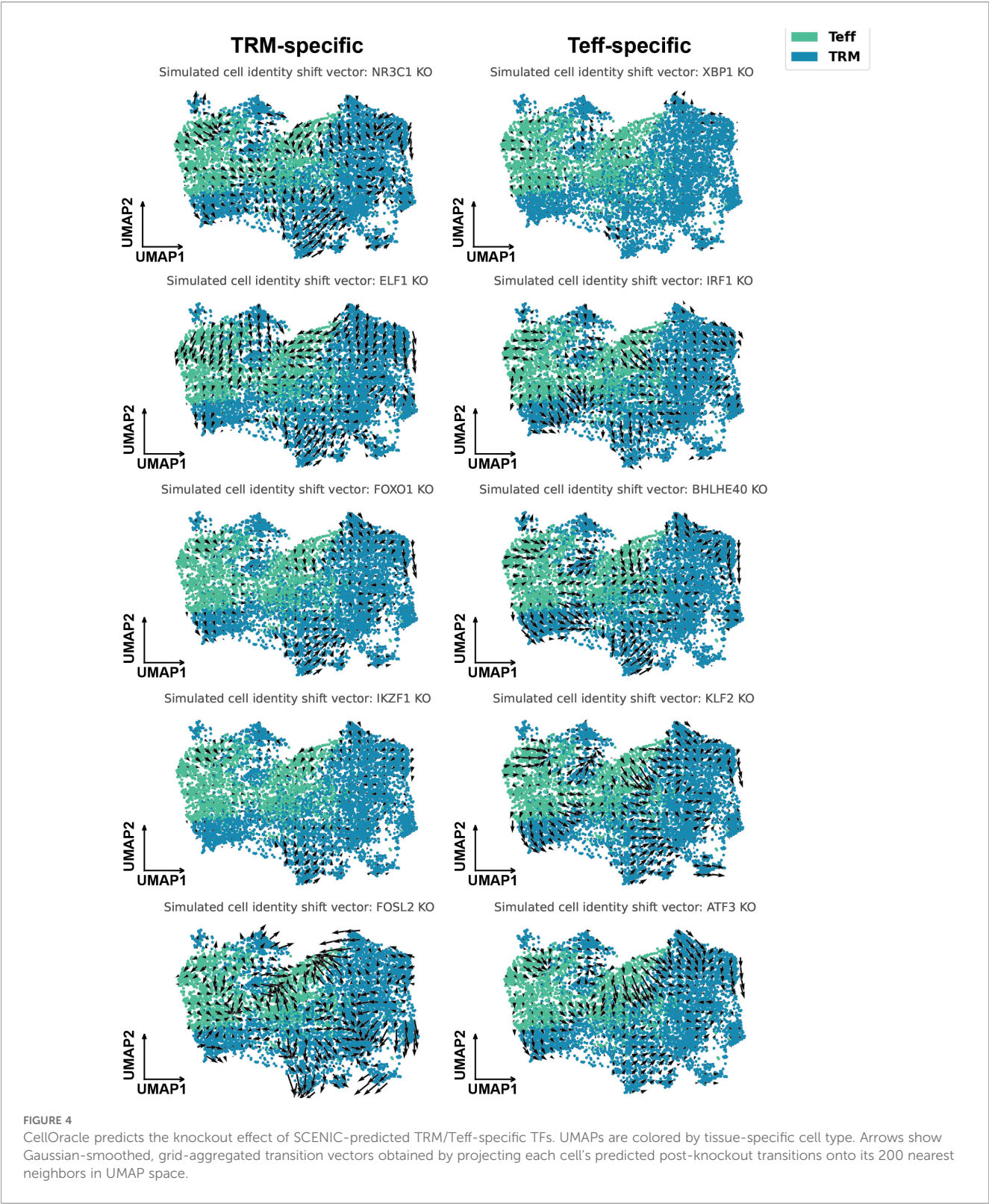


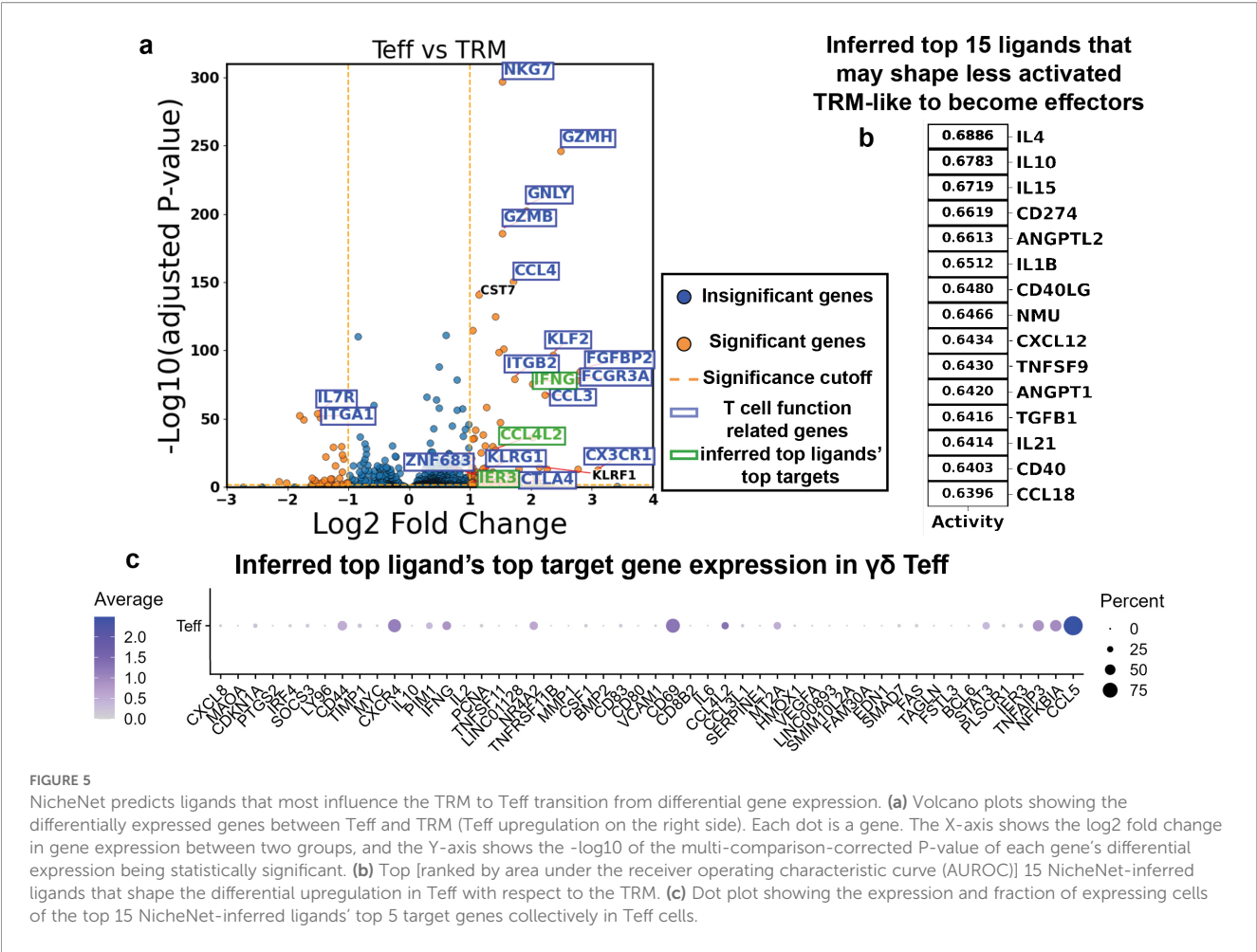
FIGURE 3

SCENIC predicts gene regulatory networks and TF activities in human CRC $\gamma\delta$ T cells. (a) pie chart showing the percentage of SCENIC predicted TF-target gene pairs that are already in the NicheNet-summarized GRN. Has Source: the SCENIC-identified pair is in the NicheNet-summarized GRN version "21122021," No source: the SCENIC-identified pair is not in the NicheNet-summarized GRN version "21122021." (b) (Left) Workflow of using ChIP-seq data to validate SCENIC prediction that is not documented in NicheNet-summarized GRN version "21122021". Overlap between a TF's target genes TSS surroundings and that TF's ChIP-seq peaks, if contained in the database, is calculated. (Right) Heatmap visualizing the percentage of TF's SCENIC-predicted target genes can be found to overlap with a peak. (c) The regulon specificity score plot showing how the collective expression of a TF's target genes converges with the cell type labels, or in other words, how specific a TF's regulatory activities are to the cell type. The x-axis is the ranked regulons.

also from MSS patients, were enriched for ligands associated with a shift toward exhaustion (Figure 7a; detailed expression heatmaps in Supplementary Figures S4a, b). In both MSS and microsatellite instability-high (MSI-H) CRC, monocytes/classical dendritic cells (McDCs) exhibited the strongest potential to drive $\gamma\delta$ T cell transition

from a less activated, TRM-like state to a more activated, effector phenotype (Figure 7b; expression heatmap in Supplementary Figure S5a). Additionally, enrichment analysis indicated that CRC fibroblasts may contribute to the exhaustion of tumor-infiltrating $\gamma\delta$ T cells (Figure 7b; detailed ligand expression in Supplementary Figure S5b).



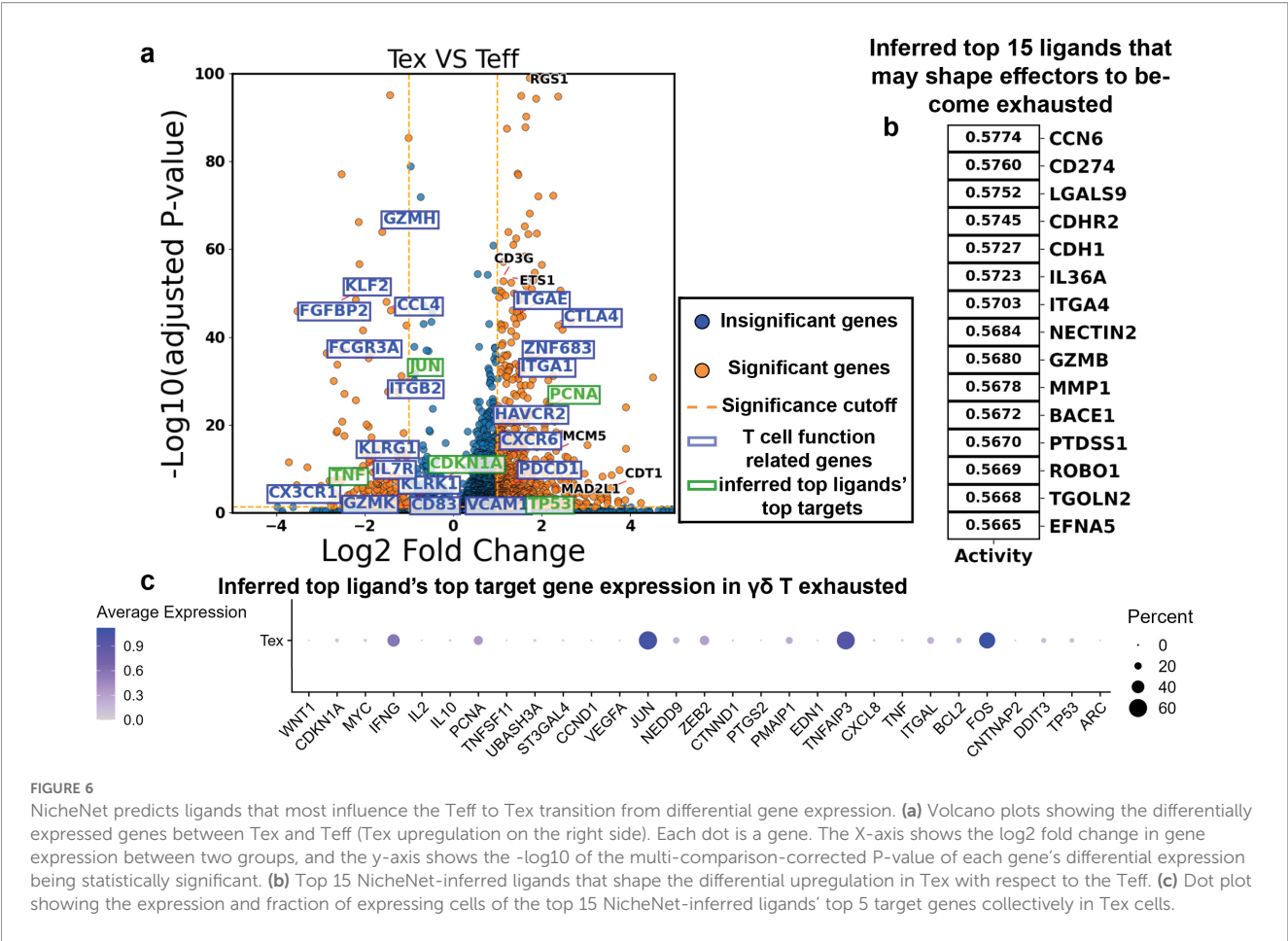


profiles. The *in silico* overexpression analysis involves simulating the increased expression of candidate genes within the original TRM cell state and assessing which gene perturbations produce a cell embedding that is similar to the embedding of the goal state, in this case, Teff (Figure 2).

Using the fine-tuned Geneformer model, which achieved 88% accuracy in classifying TRM and Teff labels (Supplementary Figure S6), we identified several genes whose *in silico* overexpression was associated with a shift in embedding toward an effector-like state. These included *CCR9*, *NCR2*, *KLRK1* (NKG2D), *KLRC3* (NKG2E), *NR4A3*, and *MMP9* (Figure 8a; full list of DEGs in Supplementary Table S8). Notably, most of these genes were not differentially expressed between Teff and TRM-like cells, indicating that Geneformer uncovered novel perturbation targets rather than simply retrieving genes whose differential expression reflects the Teff-like transcriptional profile. We then re-applied the $\gamma\delta$ T cell-refined NicheNet to the set of Geneformer-predicted perturbation genes to infer upstream ligands. Among the top predicted ligands were IL-15 and TNFSF9, which also ranked highly in the initial NicheNet analysis based on differentially expressed gene signatures (Figure 8b; full list of ligands in Supplementary Table S9).

Discussion

In this study, we developed a $\gamma\delta$ T cell-tailored ligand inference framework to investigate the molecular drivers of $\gamma\delta$ T cell activation in CRC. In our previous study, by integrating multi-study single-cell RNA-seq data, we constructed a comprehensive atlas of CRC-infiltrating $\gamma\delta$ T cells and identified major subsets, including TRM-like bystanders and cytotoxic effectors. To uncover signals responsible for driving transitions between these phenotypes, we combined differential gene expression analysis, SCENIC-inferred gene regulatory networks, NicheNet ligand inference, and the large language model Geneformer for *in silico* perturbation. We refined the NicheNet framework by incorporating $\gamma\delta$ T cell-specific ligand-receptor interactions and transcriptional regulatory relations. This approach revealed key ligands, including IL-15 and TNFSF9, as potential regulators of $\gamma\delta$ T cell activation. Notably, Geneformer identified perturbation targets such as *NCR2* and *KLRC3* (NKG2E), which were not differentially expressed but capable of inducing a Teff-like state, highlighting the importance of natural killer receptor signaling other than the well-known NKG2D and *NCR3*. Together, our framework provides a systems-level perspective on the extrinsic and intrinsic factors shaping $\gamma\delta$ T cell



phenotypes in CRC and offers a prioritized list of experimentally testable candidates for immunomodulation.

SCENIC revealed distinct TFs that regulate $\gamma\delta$ T cell subtypes in human CRC. MYBL2, associated with cell proliferation (40), may indicate a proliferative potential in $\gamma\delta$ T cells expressing exhaustion markers. SPI1 acts downstream of PRDM1 to enhance PD-L1 transcription (41), indicating a regulatory role in the Tex $\gamma\delta$ T cell subset. The CellOracle in silico perturbation further supports SCENIC's prediction that different CRC-infiltrating $\gamma\delta$ T cell states are regulated by distinct TFs. TRM is predicted to be most specifically regulated by NR3C1, ELF1, FOXO1, IKZF1, and FOSL2. Teff is predicted to be regulated by XBP1, IRF1, BHLHE40, KLF2, and ATF3. Indeed, literature-reported functions of these subset-specific TFs align well with the observed activation states. TFs enriched in the less-activated TRM subset were associated with promoting cell survival, preventing exhaustion, and suppressing effector function, whereas TFs enriched in the effector subset were primarily involved in active interferon production and signaling and in stress responses. NR3C1 encodes the glucocorticoid receptor (GR), a ligand-activated TF that mediates cortisol's immunosuppressive effects. FOXO1 maintains long-lived cells; AKT phosphorylates FOXO1, excluding it from the nucleus and thereby relieving its restraint on effector differentiation, which is consistent with the higher IL7R levels we observe in TRM (42). A study combining mouse transcriptomics and epigenomics reported that FOXO1 is required to sustain TRM in the small-intestinal lamina propria and in the colon (43). FOXO1 also antagonizes IL-17 programs, which may explain our earlier observation that human CRC $\gamma\delta$ T cells do not produce IL-17 (44). IKZF1(Ikaros), previously shown to repress gene programs associated with type I interferon responses (45), may help explain the limited effector function observed in TRM-like $\gamma\delta$ T cells. For FOSL2, chronic-stimulation experiments in mice indicate that miR-155 promotes T cell exhaustion in part by downregulating FOSL2, which then reduces AP-1 activity and permits an "exhausted" gene program (46). KLF2, whose downregulation is required for the establishment of tissue residency (47), promotes S1PR1 expression, facilitating T cell circulation through the bloodstream and lymphatic system (48–50). It has also been implicated in enhancing effector function and preventing exhaustion in CD8⁺ T cells during LCMV infection (51), raising the possibility that KLF2 may influence effector function or exhaustion dynamics in $\gamma\delta$ T cells. IRF1 is required for optimal granzyme B and perforin expression and for responsiveness to IL-12/IL-18 signaling in CD8 T cells (52). BHLHE40 responds to activation and inflammation, and human T cells edited to lack BHLHE40 show increased IL-10 and reduced IFN- γ , whereas BHLHE40 overexpression boosts IFN- γ secretion (53).

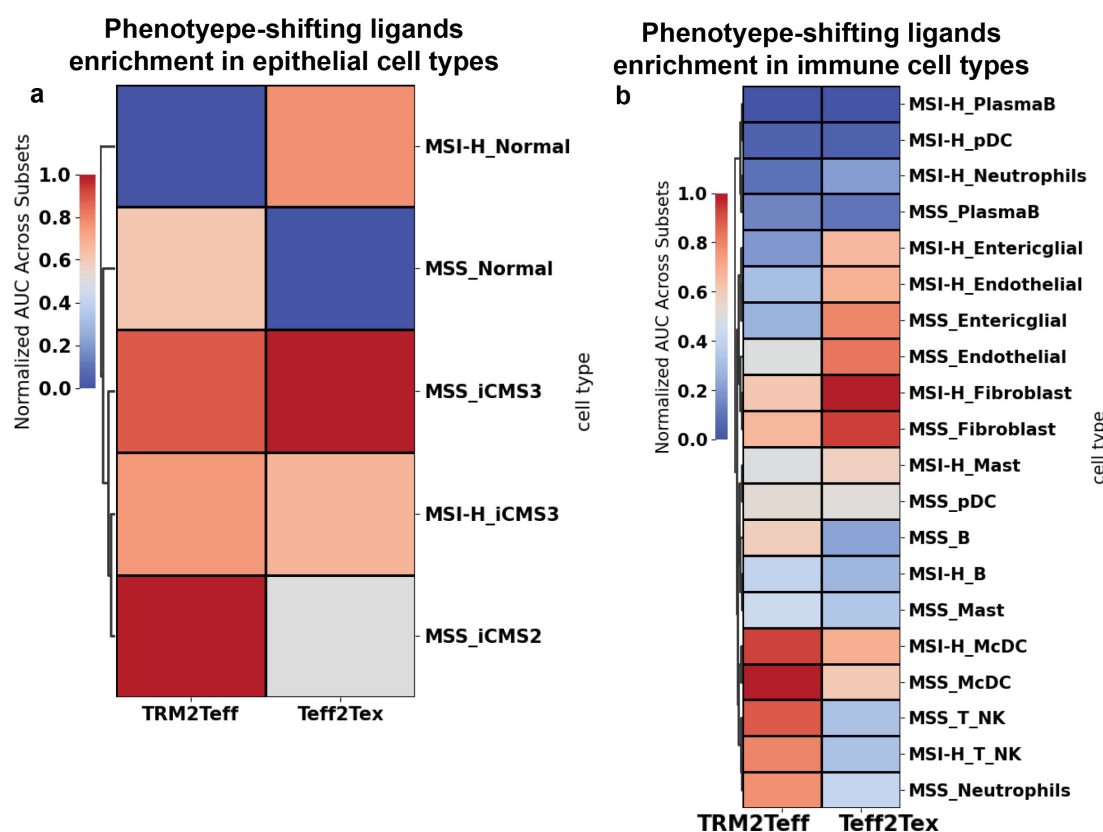


FIGURE 7

Human CRC $\gamma\delta$ T cells phenotype-shifting ligands are enriched in macrophage, dendritic cells, fibroblasts, and iCMS2/3 CRC cells. The enrichment of the NicheNet-inferred phenotype-shifting ligands in each CRC TME (a) epithelial (b) non-epithelial cell type was quantified by the area under the recovery curve (AUC) against the top 30 (ranked by area under the receiver operating characteristic curve (AUROC), AUROC > 0.5) NicheNet-inferred ligands. Results are visualized by a heatmap, where the color represents the enrichment min-max normalized across cell types. TRM2Teff, TRM to Teff. Teff2Tex, Teff to Tex. MSS, microsatellite stable; MSI-H, microsatellite instability-high.

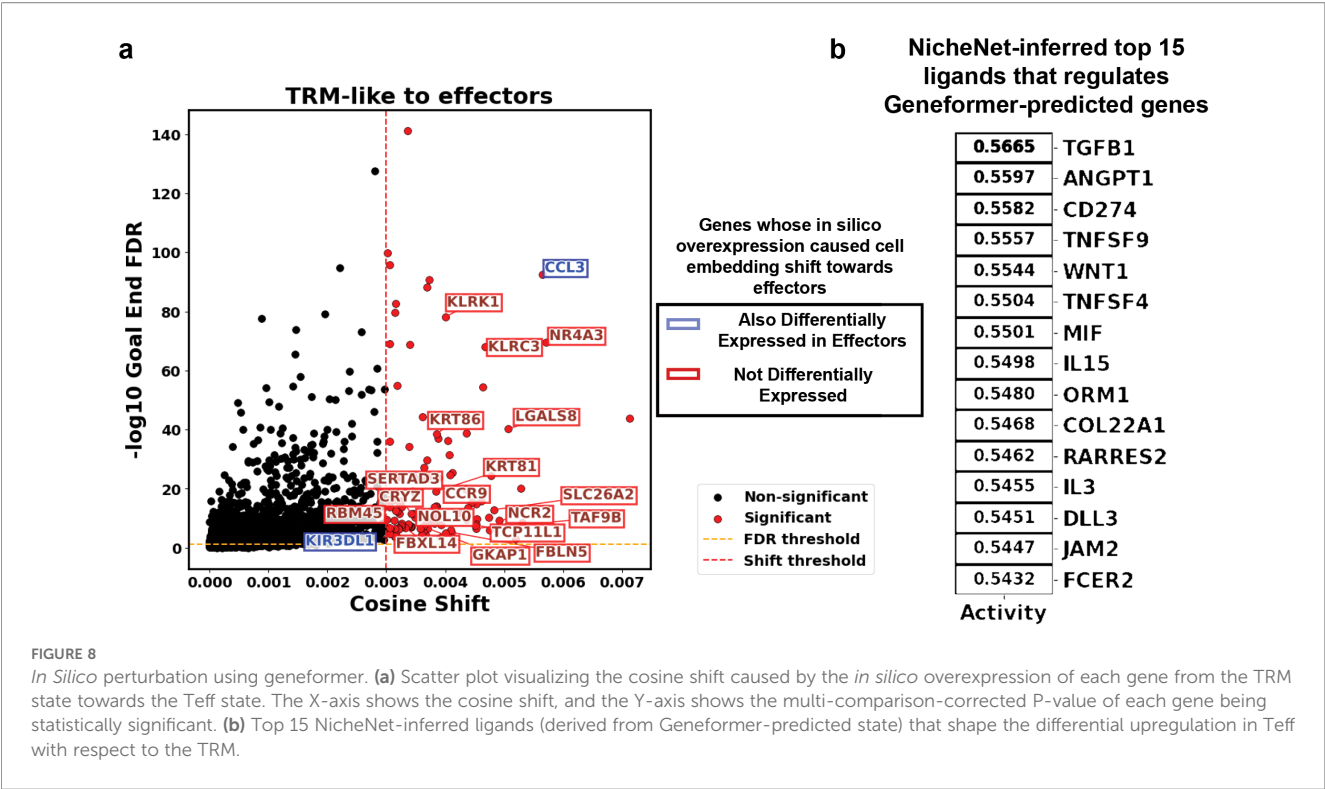
Our DEG analysis highlights key features of $\gamma\delta$ T effector cells in human CRC. *KLRG1*, a marker of terminal differentiation and senescence in CD8⁺ effector T cells (54, 55), was highly expressed in $\gamma\delta$ Teff cells in our integrated dataset. As an inhibitory receptor, *KLRG1* impairs T cell effector function by blocking AKT phosphorylation through the recruitment of phosphatases that convert PIP3 to PIP2 (56). In murine models, *KLRG1* inhibition enhances antitumor activity (57). Flow cytometry studies in human colorectal cancer have shown that *KLRG1*⁺ T cells exhibit impaired cytotoxicity (58); however, the functional consequences of *KLRG1* expression in $\gamma\delta$ T cells remain unclear. It is yet to be determined whether *KLRG1* dampens $\gamma\delta$ T cell function as it does in natural killer (NK) cells (59), or whether its expression is more permissive, as has been debated for PD-1, which may not signify exhaustion in $\gamma\delta$ T cells but instead reflects an activation state (60).

Interestingly, *KLRG1*⁺ CD8⁺ Teff cells have been shown to downregulate *KLRG1* and differentiate into memory subsets (61). *KLRG1*⁺ $\gamma\delta$ T cells are also found in CMV-uninfected newborns (62) and become more prominent following CMV infection and in adulthood (63). In murine cytomegalovirus (MCMV) infection, *KLRG1*⁺*CX3CR1*⁺ $\gamma\delta$ effector memory cells populate lung and liver vasculature and mount strong antiviral responses (64). Although a

scRNA-seq analysis of CRC patient samples from TCGA reported enrichment of *KLRG1*⁺ cytotoxic T cell signatures associated with favorable outcomes (65), it may be premature to attribute a positive role to *KLRG1* in antitumor immunity.

Additionally, *ZNF683* (Hobit), which has been implicated in promoting IFNG transcription in effector T cells (66) and marks precursors of tissue-resident memory T cells (67), was also upregulated in $\gamma\delta$ Teff cells, supporting their cytotoxic potential and tissue-localized phenotype. Compared to the less activated TRM-like population, Teff cells also showed downregulation of *IL7R* and *ITGA1*, suggesting reduced emphasis on survival and increased migratory capacity.

Ligand prediction by NicheNet in the TRM-Teff transition provided many interesting targets for $\gamma\delta$ TCR-independent activation. IL15, needless to say, is a well-studied ligand essential for T cell survival, proliferation, and effector function (68). It has been used in multiple studies to expand V δ 2⁺ $\gamma\delta$ T cells or CAR $\gamma\delta$ T cells and enhance their cytotoxicity (69–71), although there are also reports that continuous IL-15 signaling leads to exhaustion in NK cells (72, 73). Another interleukin in the list, IL-21, is also known to promote T cell proliferation and cytotoxic differentiation (74). Literature has reported its effect on providing short-term



proliferation (75) and enhancing the antitumor cytotoxic response in $\gamma\delta$ T cells in the presence of IL-2 (76). Studies have also shown IL-21's synergistic effect with IL-15 in promoting CAR T and NK cell expansion (77, 78). Interestingly, both CD40 and its ligand, CD40LG, were among the top ligand list. Recent findings show that CD40LG expressed on CD8⁺ T cells can engage CD40 on dendritic cells, leading to their activation and promoting CD8⁺ T cell effector differentiation (79). The CD40LG on CD8 T cells is found to interact with CD40 on dendritic cells, such that this interaction not only promotes CD8 T cells' effector function (80) but also the survival of dendritic cells in antitumor response (81). In mice parasite infection, $\gamma\delta$ T cells enhance dendritic cells activation through CD40LG-CD40 interactions (82). Considering that TRM-to-Teff NicheNet-predicted ligands were most enriched in McDCs in Joanito et al.'s dataset, a model where $\gamma\delta$ T cells promote dendritic cell activation via CD40LG-CD40 interactions, thereby reinforcing their own effector programming, is a reasonable hypothesis. Indeed, although $\gamma\delta$ cells are known for MHC-independent activation, our result of McDCs being identified as the top candidate influencing $\gamma\delta$ cells' effector differentiation encourages future studies that focus on antigen-presenting cells (APC)- $\gamma\delta$ cells interactions.

On the other hand, there is also a possibility that CD40 is expressed on $\gamma\delta$ T cells and interacts with CD40LG⁺ cells in the TME. The expression of CD40 on activated CD8⁺ T cells has been reported (83, 84), and *in vitro* stimulation of CD40⁺ human CD8⁺ T cells with CD40LG⁺ artificial APCs increased both the total number and percentage of effector cells (84). Although such CD40 expression appears to be transient, it may still exert long-term

effects on T cell differentiation, at least in terms of cell numbers, as pointed out in a previous study (84).

Another interesting finding is the presence of TNFSF9, the 4-1BB ligand, among the top Teff-shifting ligands, rather than CD80/86, which bind CD28. In T cells, TNFSF9 provides a potent co-stimulatory signal through 4-1BB, amplifying the proliferation and survival of effector T cells (85). Since the second generation of CAR-T therapy, the intracellular domain of 4-1BB has been incorporated into the CAR design (86) to achieve a better activation. Current CAR $\gamma\delta$ T cells typically use V δ 2 TCRs, the most abundant $\gamma\delta$ T subset in blood, and incorporate the CD28 (87) signaling domain based on second-generation CAR $\alpha\beta$ T cell design (14). However, whether CD28 effectively helps the activation of most intraepithelial $\gamma\delta$ T cells, or generally speaking, the V δ 1 subset, which has stronger cytotoxicity (88), tissue residency (89), and tumor infiltration (90–92), remains unclear. Our preliminary scRNA expression data of intraepithelial $\gamma\delta$ T cells in healthy human colon (93) agrees with the previous mouse (94) and human (95) observations that they do not express CD28. Additionally, there are contradictory results of the CD28 antibody's effect on human $\gamma\delta$ T cells (96, 97). There are multiple potential therapeutic co-stimulatory receptors, like 4-1BB (98), CD27 (99), and CD6 (100), but stimulating them in the CRC settings has unknown results, plus there could be more undiscovered. The identification of 4-1BBL among the top ligands but not CD80/86 suggests that 4-1BB is likely to serve as a more prominent or contextually relevant co-stimulatory signal for $\gamma\delta$ T cells in human CRC. Indeed, ligand prediction based on Geneformer-derived overexpression targets also supported the importance of 4-1BBL.

The Geneformer-identified perturbation targets are also very interesting. The NKG2D and NCR3 have been relatively well discussed in $\gamma\delta$ T cells, but our analysis also highlights the NCR2 and NKG2E. Unlike conventional $\alpha\beta$ T cells, $\gamma\delta$ T cells can upregulate NCR2 (NKP44) upon activation. Resting peripheral $\gamma\delta$ T cells lack NKP44, but cytokine-driven stimulation induces its expression in a subset of cells (101). Notably, NKP44 induction is largely restricted to $V\delta 1^+$ $\gamma\delta$ T cells. Prolonged TCR stimulation (more than 2 weeks) in the presence of IL-15 or IL-2 leads to robust NKP44 (and NKP30) expression in $V\delta 1^+$ cells, whereas $V\delta 2^+$ cells show minimal NKP44 upregulation (102, 103). KLRC3, alias NKG2E, is an activating NK receptor that is closely related to the well-known NKG2C. Both NKG2C and NKG2E form heterodimers with CD94 and signal through the DAP12 adapter, analogous to NK cells (104). CD94-NKG2C recognizes HLA-E and signals through the ITAM adaptor DAP12, recruiting Syk/ZAP-70 to promote cytotoxicity (105).

Admittedly, our pipeline relies on the sequential application of several tools (SCENIC, NicheNet, and Geneformer), which may introduce compounded modeling uncertainty. We use these tools not as definitive sources of truth but as complementary, mechanistically interpretable aids for prioritizing hypotheses, especially in the context of rare cell types such as $\gamma\delta$ T cells, where validated PPI and TF-target data are scarce. In this way, inputting predictions from one tool into another represents a practical way to take cell type specificity into consideration in a time- and cost-efficient manner. Nonetheless, we expect that as $\gamma\delta$ T cell-specific interaction data are experimentally validated, our framework can be progressively updated and benchmarked against these ground truths. Likewise, if ChIP-seq data for all TFs predicted by SCENIC become available in CRC $\gamma\delta$ T cells, the verification shown in Figure 3, which currently relies on ChIP-seq data from general T cells and NK cells in public databases due to resource limitations, would be more convincing. An additional limitation is that we did not yet explicitly account for the heterogeneity in sex, age, and anatomical location present in the integrated single-cell datasets, and future extensions incorporating bootstrapping and subgroup modeling will be required to systematically evaluate its impact on $\gamma\delta$ T cell activation states and ligand associations.

In this current study, we focused on genes upregulated in Teff versus TRM as the NicheNet gene set of interest and on overexpression-like perturbations from Geneformer. This activation-centric design matches the default pySCENIC output of positive regulons and mitigates some of the ambiguity introduced by dropout when interpreting downregulation in scRNA-seq data. The use of positive DEGs as the input for NicheNet is also practiced by single-cell best practices (106). Nonetheless, future extensions of our framework could explicitly incorporate downregulated genes and knockdown-like perturbations to identify ligands that stabilize TRM-like programs or repress effector differentiation.

For ligand inference, several other tools are also available. For example, scMLnet (107) reconstructs a multilayer signaling network by integrating prior knowledge with single-cell expression data. It first defines a ligand-receptor subnetwork by selecting highly expressed ligands in the sender population and cognate receptors

in the receiver population, then builds a TF-target subnetwork in the receiver by testing enrichment of known TF-target relationships among highly expressed genes to infer activated TFs. These activated TFs are then connected back to upstream receptors using curated receptor-TF interactions, and the ligand to receptor, receptor to TF and TF to target subnetworks are merged into a cascade that is further refined using receptor-TF and TF-target expression correlations. Similarly, scSeqComm (108) combines statistical and network-based steps: it computes an intercellular signaling score for each candidate ligand-receptor pair from cluster-level expression, and then an intracellular signaling score that quantifies how strongly downstream signaling pathways and TF regulons linked to the receptor are activated in the receiver cells, based on curated signaling-pathway and transcriptional-regulatory networks. Conceptually, both scMLnet and scSeqComm share NicheNet's reliance on curated ligand-receptor and regulatory knowledge, but they differ in how they score ligand-target connections: NicheNet propagates ligand signals over an integrated prior network using a Personalized PageRank-based scheme, whereas scMLnet and scSeqComm explicitly prioritize the ligand-target chain using dataset-specific statistical evidence.

To validate the predicted ligand-receptor interactions, we propose complementary spatial and perturbation approaches. Ideally, spatial transcriptomics and multiplexed protein imaging would be used to confirm that $\gamma\delta$ T cells and ligand-expressing cells are in direct spatial proximity. As a computational alternative, $\gamma\delta$ T cell transcriptional profiles can be mapped onto CRC spatial transcriptomics sections (for example, Visium) using tools such as Tangram (109). One can then quantify whether predicted ligand-expressing cells or niches are enriched for $\gamma\delta$ T cell probability using spatial cross-correlation metrics and local Moran's I, and compare the observed co-localization to null distributions generated by spatially constrained spot permutations or block randomization within tissue regions. In parallel, pooled CRISPR Perturb-seq in primary tumor-infiltrating $\gamma\delta$ T cells can be used to test causality. An sgRNA library would target receptors prioritized by this study and key downstream signaling nodes, including non-targeting and pathway-positive controls. Transduced cells would be cultured in CRC tissue-conditioned medium whose cytokine and ligand composition is quantified by a multiplex platform, then profiled by single-cell RNA-seq with guide calling and, optionally, CITE-seq antibodies for Teff, TRM, and exhaustion surface markers.

Several emerging immunotherapies, including immune checkpoint blockade, CAR T cell therapy, and bispecific T-cell engagers, are being actively explored to enhance anti-tumor immune responses. Immune checkpoint inhibitors (e.g., antibodies targeting PD-1) have achieved durable responses in the subset of CRC patients with mismatch-repair deficient, microsatellite instability-high (MSI-H) tumors, leading to regulatory approvals for this population (110). Chimeric antigen receptor (CAR) T cell therapy is also being pursued in CRC by engineering patient T cells to recognize tumor-associated antigens, and early-phase trials have reported initial safety and antitumor activity with this approach, even though CAR T strategies for CRC remain in early development (111). Another innovative strategy

involves bispecific antibodies. These agents are often formatted as bispecific T-cell engagers, and they bind simultaneously to a tumor antigen and to the CD3 complex on T cells, physically linking immune cells to cancer cells and triggering tumor cell lysis. Prototypes targeting CRC antigens like EGFR or carcinoembryonic antigen have demonstrated potent tumor cell killing in preclinical models and even in patient-derived samples (112). Recent advances in $\gamma\delta$ T cell-based immunotherapy, including CAR $\gamma\delta$ T cell applications in hematological malignancies and solid tumors, have been comprehensively reviewed by Dong et al. (14), providing important clinical and translational context for our findings.

Together, these findings highlight ligands such as IL-15 and TNFSF9 as promising candidates for promoting $\gamma\delta$ T cell activation in CRC. In addition, NCR2 and NKG2E emerged as potential receptors through which $\gamma\delta$ T cells may recognize CRC. These insights can inform the design of engineered $\gamma\delta$ T cell therapies aimed at enhancing antitumor function and supporting both early tumor elimination and long-term postsurgical surveillance.

Method

Data acquisition

The human CRC-infiltrating $\gamma\delta$ T cell scRNA-seq dataset was curated as described in our previous study (22). Briefly, we searched for publicly available scRNA-seq datasets from treatment naïve CRCs in indexed journals and Gene Expression Omnibus. In each study, cell clusters that highly co-expressed CD3D, CD3E, CD3G, CD247, and TRDC were identified as $\gamma\delta$ T cells. Distinct $\gamma\delta$ T cell clusters with more than 50 cells were identified from 9 studies that provided raw counts. Human CRC whole-tissue scRNA-seq datasets were obtained from Gene Expression Omnibus (GEO) Series GSE178341 (113), GSE200997 (114), GSE245552 (115), GSE231559 (116), GSE188711 (117), GSE161277 (118), GSE108989 (119), GSE178318 (120), and PubMed PMID35773407 (39). The datasets analyzed in this study were derived from a diverse cohort of patients, encompassing different sexes, anatomical sites, demographic backgrounds, and disease stages, as detailed in our previous work. Since this study involved secondary analysis of publicly available scRNA-seq datasets, no additional group allocation or randomization was performed by the authors. Investigators were not blinded during data analysis, as all datasets were de-identified and publicly accessible. No power calculations were conducted, as sample sizes were determined by the availability and characteristics of the original datasets. Detailed experimental protocols and patient recruitment information can be found in the original publications associated with each dataset.

Data processing

For processing the original dataset, only cells in the colon mucosa were kept. If processed data is not provided, Seurat (121) and Scanpy

(122) were used for downstream analysis. For quality control, low-quality cells were dropped based on their low UMI counts (<500), high mitochondrial gene counts (>20%), and a low number of uniquely expressed genes (<200). If the unprocessed data of a study has multiple samples, data integration was done by using Seurat on the normalized and log-transformed raw counts, which means each cell's counts are divided by its total counts, multiplied by 10,000, then log-transformed. SCTransform was applied to raw counts of $\gamma\delta$ T cells isolated from each study. As we have briefly summarized in the previous study, raw counts were fitted into negative binomial distributions whose expectation was a function of the total count of cells. The coefficients of the gene's general linear model are further regularized by the kernel regression with the coefficients of other genes that have similar average expression. Regressed gene expression calculated from the regularized generalized linear model coefficients was referred as "SCTransform-corrected counts," and log-transformed SCTransform-corrected counts were used for later visualization. The Pearson residual of a cell's observed gene expression to the SCTransform-corrected counts was used for integration and later performing principal component analysis (PCA, 50 pcs kept). Leiden clustering was performed based on the computed neighborhood graph of observations (UMAP, 50 pcs, size of neighborhood equals 15 cells) to reveal the general subtypes. In order to delineate cell subtypes, further subclustering was performed on each subcluster at a resolution range from 0 to 1 (detailed resolution for each step of subclustering was recorded in the published code). The cell type annotating strategy was detailed in our previous study. In short, we identified four distinct subtypes of $\gamma\delta$ T cells in our integrated dataset: poised effector-like T cells (Teff, markers: *KLF2*, *KLRG1*, *TBX21*, *IFNG* and low *IL7R*, *ITGAE*, *ITGA1*, *CCR7*, *SELL*), tissue-resident memory T cells (TRM, markers: *ITGAE*, *ITGA1* and low *CCR7*, *SELL*, *KLF2*), progenitor exhausted T cells (Tpex, markers: *TCF7*, *PDCD1*, *CTLA4*, *LAG3*, *TIGIT*, *HAVCR2*), and Tex (*GZMB*, *PDCD1*, *CTLA4*, *LAG3*, *TIGIT*, *HAVCR2* and no *TCF7*). Effector and exhaustion scores were calculated for each cell based on the aforementioned genes using the scanpy function `score_genes` against the same number (6) of random genes.

For data analysis, the tools we used are detailed below. We also summarize the inputs and outputs of these tools in **Supplementary Table S9** to help readers conveniently follow our workflow. SCENIC: Cell expression profile of the tumor-infiltrating $\gamma\delta$ T cells was input into the SCENIC (v1.3.1) pipeline using default parameters. List of TFs "hs_hgnc_curated_tfs.txt," motifs "motifs-v10nr_clust-nr.hgnc-m0.001-o0.0.tbl," and the motif ranking databases "hg38_500bp_up_100bp_down_full_tx_v10_clust.genes_vs_motifs.rankings.feather" used for the pipeline were obtained from <https://resources.aertslab.org/cistarget/>, http://github.com/aertslab/pySCENIC/blob/master/resources/hs_hgnc_curated_tfs.txt, and <https://resources.aertslab.org/cistarget/motif2tf/>. As indicated in the file name, our motif search range is 500 bp upstream and 100 bp downstream of the target gene TSS. The output regulons from the cisTarget are integrated into the NicheNet's network. Briefly, these regulons are considered statistically significant. SCENIC calculates the enrichment of a TF's motif in its potential target

genes from their gene vs. motifs ranking matrix. First, by default, cisTarget keeps the top 75% of the regulon members (target genes) of a motif by the motif's enrichment in the gene's promoter region. Second, a recovery curve for the motif is established by the percentage of inferred target genes recovered from parsing down the ranked list. This area under the curve (AUC) will be normalized over the AUC of all motifs to recover the same set of target genes (i.e., the inferred target genes of the motif) from all genes to obtain the Normalized Enrichment Score (NES), and a motif with NES > 3 by default is considered significant to regulate the inferred target set. For the ChIP-seq verification of the SCENIC prediction, the chromosome coordinates were obtained from Ensembl "Homo_sapiens.GRCh38.113.chr.gtf.gz." The ChIP-seq data were obtained from ChIP-atlas' "Homo Sapiens ChIP: TFs and others" category, with "T cells," "Th1 Cells," "Th17 Cells," "CD8+ T cells," "memory T cells," "Natural Killer T cells," "Th2 Cells," "CD4+ T cells," and "Natural Killer cells" are being selected as cell types. GenomicRanges was used to simply find the overlap between the +/- 5kb area around the target gene TSS and the peak of the TFs, which are in both SCENIC predictions and the ChIP database. The percentage of a TF's target genes that could be found at least 1 peak of that TF around its TSS in all obtained ChIP-seq data was visualized. Regulon specificity score of TF in cell subsets is calculated using the pySCENIC function "plot_rss." Briefly, in SCENIC, the regulon specificity score quantifies how specifically a TF regulon is active in a given cell type. Briefly, SCENIC first ranks gene expression in each cell and uses AUCCell to compute an AUC score per regulon per cell, reflecting how well the regulon's target genes are recovered in that cell. For a given cell type X, two vectors are then defined: P, the distribution of AUC values for that TF's regulon across all cells, and Q, a binary vector indicating whether each cell belongs to cell type X or not. The RSS is defined as $1 - JS(P||Q)$, where JS denotes the Jensen-Shannon divergence to describes the divergence of 2 distribution. Higher RSS values indicate that the regulon's activity is more specifically enriched in that cell type.

CellOracle in silico perturbation

We used the prebuilt promoter-based human base GRN hg19_gimmotifsv5_fpr2 and the authors' recommended automated threshold (123) to construct GRN models, then performed *in silico* KO with default settings (perturbation: set TF expression to 0; n_propagation = 3 to compute post-KO expression; n_neighbors = 200 to estimate transition probabilities). For visualization, all quiver/vector-field plots were drawn with a common scale.

NicheNet

NicheNet (25) (v2.2.0) is used for ligand inference. The integrated ligand-receptor pairs, signaling network, and gene regulatory networks (prefix 21122021) were obtained from the package's repository. TF-target gene pairs that are not in

NicheNet's original network and $\gamma\delta$ T cell-specific ligand-receptor pairs (CD247 and its ligands EPHA2, EPCR, BTN2A1, BTNL3, BTNL8, CD1B, CD1D, MR1, CD1A, CD1C) were added to the networks with unoptimized weights of 1, given the relatively small number of edges being added and the original author's comments on the robustness of NicheNet without weight optimization. Functions "construct_weighted_networks," "apply_hub_corrections," and "construct_ligand_target_matrix" from the NicheNet package were used sequentially to construct the new ligand-target matrix following NicheNet's user guide with default parameters. The hyperparameters used during construction were the same as those of the unmodified NicheNet, which was obtained from the package's repository under the name "hyperparameter_list.rda." Function "evaluate_model" from the NicheNet package was used to evaluate both the unmodified ligand-target matrix and the new ligand-target matrix's performance on the "gold standard" dataset that the NicheNet authors used.

The application of the NicheNet is similar to and adapted from our previous study (93). NicheNet was supplied with DEGs and Geneformer. DEGs were calculated by using Scanpy's tl.rank_genes_groups, Wilcoxon method. DEGs with adjusted p-values < 0.05 and log₂ fold changes greater than 1 were used as input for NicheNet. Genes expressed in at least 1% of the cells in the receiver subsets (goal states, Teff in the TRM to Teff transition, and Tex in the Teff to Tex transition) were considered as background. Potential ligands were defined as ligands of NicheNet-documented receptors whose encoding genes were expressed in at least 1% of the cells. The top 15 ligands with an AUROC > 0.5 were reported in the main figure, while the top 30 were used for quantifying their enrichment in neighboring cell types. If fewer than 15 ligands met the AUROC > 0.5 threshold, all qualifying ligands were visualized and included in the enrichment analysis.

CRC TME cell type profiles were obtained from Joanito et al. (39). The datasets, epithelial and nonepithelial, had already been quality controlled by the providers. Basic pre-processing, including normalization and log transformation, was done by using Scanpy's normalize_total and log1p functions. Cells labeled "LymphNode" and "Normal" were excluded. AUC calculations were performed after averaging gene expression by cell type, which is annotated by the original authors. A gene was included in the averaged profile only if its average expression was not less than 0.001 to reduce noise. For each cell type, we iterated down the ranked gene list (ranked by their average expression in the given cell type, descending order) to recover NicheNet-inferred ligands, stopping when encountering a zero-expression gene. The final area under the curve was computed using the auc function from sklearn.metrics.

Geneformer

Geneformer (26) (v0.1.0) was used to perform *in silico* overexpression. Data that only contained TRM and Teff was tokenized using Geneformer's TranscriptomeTokenizer, retaining protein-coding and miRNA genes by default. The pre-trained model "gf-12L-95M-i4096" was fine-tuned by an input data cell type label

classification task, using the function “Classifier.validate” with 100 hyperparameter optimization trials, with a train-valid-test split of 0.6-0.2-0.2, max_ncells=None, freeze_layers = 6. The model with the highest accuracy was selected, and its performance as a classifier was visualized using the function “plot_conf_mat.” The initial state (TRM) and the goal state (Teff) were represented by the exact mean [CLS] token embeddings of cells in each group. In silico overexpression was performed using the function “InSilicoPerturber” with max_ncells=5000 and emb_layer=-1. The statistical test of the perturbation result is computed using the function “InSilicoPerturberStats” under the mode “goal_state_shift.” Briefly, in Geneformer, significance was assessed using Wilcoxon rank-sum tests, comparing shifts induced by each gene against the background distribution of all perturbations, with Benjamini-Hochberg correction. Genes with adjusted $p < 0.05$ are considered statistically significant, and a mean cosine shift > 0.003 (empirically selected to reduce background noise) were prioritized. Genes with more than 50 detections in the initial state cells ($N_{\text{Detections}} > 50$) were reported.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: <https://www.ncbi.nlm.nih.gov/geo/>, GSE178341 <https://www.ncbi.nlm.nih.gov/geo/>, GSE200997 <https://www.ncbi.nlm.nih.gov/geo/>, GSE245552 <https://www.ncbi.nlm.nih.gov/geo/>, GSE231559 <https://www.ncbi.nlm.nih.gov/geo/>, GSE188711 <https://www.ncbi.nlm.nih.gov/geo/>, GSE161277 <https://www.ncbi.nlm.nih.gov/geo/>, GSE108989 <https://www.ncbi.nlm.nih.gov/geo/>, GSE178318 <https://pubmed.ncbi.nlm.nih.gov/>, PMID35773407.

Ethics statement

Ethical approval was not required for the study involving humans in accordance with the local legislation and institutional requirements. Written informed consent to participate in this study was not required from the participants or the participants’ legal guardians/next of kin in accordance with the national legislation and the institutional requirements.

Author contributions

RR: Writing – original draft, Writing – review & editing. DB: Writing – review & editing.

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Conflict of interest

The authors declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fimmu.2025.1715827/full#supplementary-material>

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